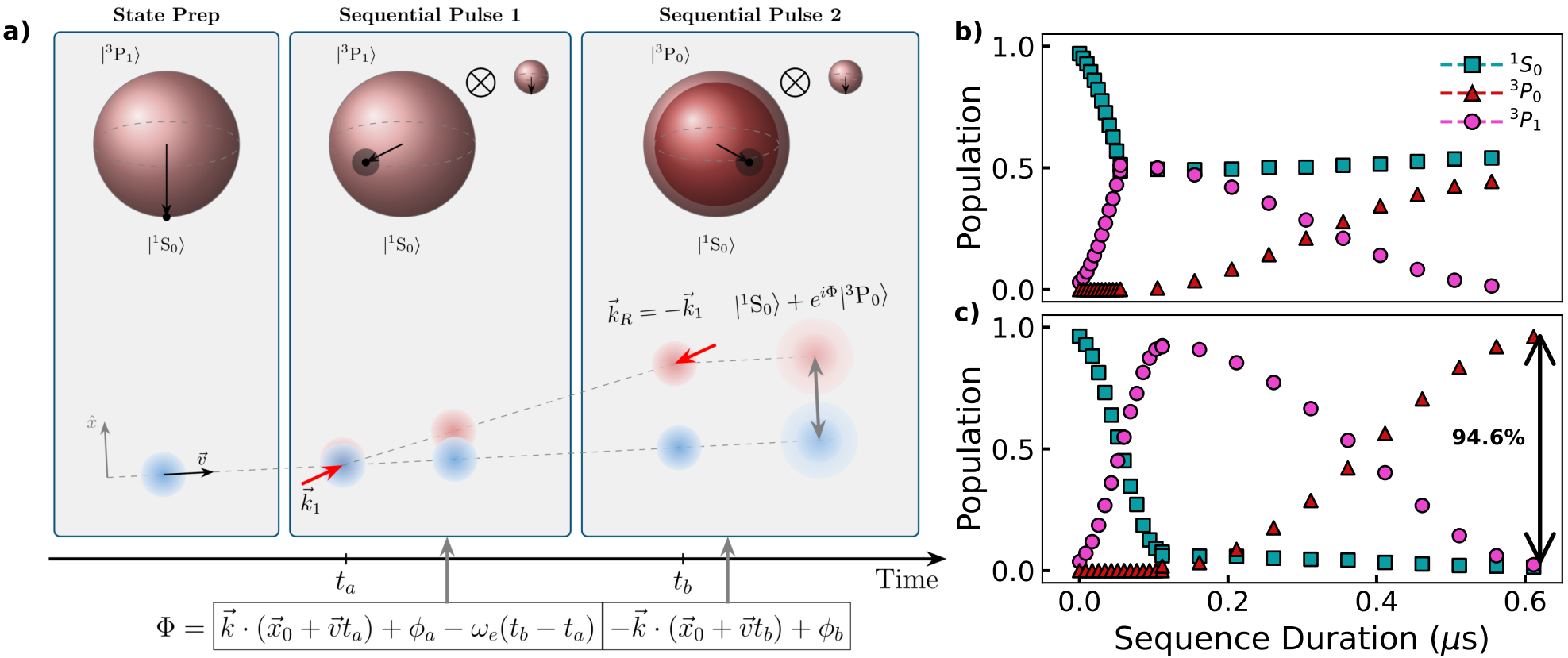
a)
State Prep
Sequential Pulse 1
Sequential Pulse 2
$|^3P_1\rangle$
$|^1S_0\rangle$
$|^3P_1\rangle$
$|^1S_0\rangle$
$|^3P_0\rangle$
$|^1S_0\rangle$
$\vec{k}_R = -\vec{k}_1$
$|^1S_0\rangle + e^{i\Phi}|^3P_0\rangle$
$\hat{x}$
$\vec{v}$
$\vec{k}_1$
t_a
t_b
Time
$\Phi = \vec{k}\cdot(\vec{x}_0 + \vec{v}t_a) + \phi_a - \omega_e(t_b - t_a)$
$-\vec{k}\cdot(\vec{x}_0 + \vec{v}t_b) + \phi_b$
b)
c)
Population
1.0
0.5
0.0
$^1S_0$
$^3P_0$
$^3P_1$
94.6%
0.0
0.2
0.4
0.6
Sequence Duration (μs)